Bootstrap:

Bootstrap is the popular HTML, CSS and JavaScript framework (A framework is a library that you can use to build software. It serves as a foundation (base), so you're not starting entirely from scratch.) for developing a responsive and mobile friendly website.

Features of Bootstrap:

- It is a front-end framework used for easier and faster web development.
- It includes HTML and CSS based design templates for typography, forms, buttons, tables, navigation, modals, image carousels and many others.
- It can also use JavaScript (is a scripting language that enables you to create dynamically update content, control multimedia, animate images, and client side validations) plug-ins.
- It facilitates you to create responsive websites which are compatible over varieties of hardware platforms like mobile phone, desktops, laptops, tablet PC's, ipad etc.
- It is absolutely free to download and use.

History of Bootstrap:

Bootstrap was developed by **Mark Otto and Jacob Thornton** at Twitter. It was released as **an open source product** in August 2011 on GitHub.

In June 2014 Bootstrap was the No.1 project on GitHub.

Why to use Bootstrap?

Following are the main advantage of Bootstrap:

- It is very easy to use. Anybody having basic knowledge of HTML and CSS can use Bootstrap.
- It facilitates users to develop a responsive website (A website is called responsive website which can automatically adjust itself to look good on all devices, from smartphones to desktops etc.).
- It is compatible on most browsers like Chrome, Firefox, Internet Explorer, Safari and Opera etc.

What Bootstrap framework (package) contains?

1) Bootstrap provides a basic structure of a web page with Grid System, link styles, and background etc.

2) Bootstrap comes with the feature of global CSS settings, fundamental HTML elements style and an advanced grid system.

3) Bootstrap contains a lot of reusable components built to provide iconography, dropdowns, navigation, alerts, pop-up, and much more.

4) Bootstrap also contains a lot of custom jQuery (jQuery is a lightweight, "write less, do more", JavaScript library. The purpose of jQuery is to make it much easier to use JavaScript on your website.) plugins which are developed in Javascript . You can easily include them all, or one by one in web page development.

5) Bootstrap components are customizable and you can customize Bootstrap's components, LESS variables, and jQuery plugins to get your own style.

Note:

While developing a website using bootstrap, we just want to use predefined HTML, CSS, Javascript templates in our program.

As we know, bootstrap framework is open source hence we can easily download it from site https://www.getbootstrap.com/

Here on this web site different versions of bootstrap are available, we can download as per our requirement. Note that the latest version right now is 5.3 (as on Oct-2023). The latest version contains all features of previous versions and also adds extra features which are not found in previous versions.

Bootstrap framework hierarchy is shown as-

```
bootstrap/
├── css/
│   ├── bootstrap.css
│   ├── bootstrap.css.map
│   ├── bootstrap.min.css
│   ├── bootstrap.min.css.map
│   ├── bootstrap-theme.css
│   ├── bootstrap-theme.css.map
│   ├── bootstrap-theme.min.css
│   └── bootstrap-theme.min.css.map
├── js/
│   ├── bootstrap.js
│   └── bootstrap.min.js
└── fonts/
    ├── glyphicons-halflings-regular.eot
    ├── glyphicons-halflings-regular.svg
    ├── glyphicons-halflings-regular.ttf
    ├── glyphicons-halflings-regular.woff
    └── glyphicons-halflings-regular.woff2
```

How to Use Bootstrap framework?

There are two ways to use bootstrap viz-

1) **Online: (BootstrapCDN (Content Delivery Network))**

   We can use the bootstrap library to develop our web site which is hosted on the internet called BootstrapCDN. Basically, BootstrapCDN contains precompiled CSS and JavaScript files.

❖ To use precompiled CSS files, we just have to use <link> tag **under <head>** tag by following way-

<link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css" rel="stylesheet" integrity="sha384-T3c6CoIi6uLrA9TneNEoa7RxnatzjcDSCmG1MXxSR1GAsXEV/Dwwykc2MPK8M2HN" crossorigin="anonymous">

Note: we can copy the code of above <link> tag from www.getbootstrap.com

❖ To use precompiled Javascript files, we just have to use <script> tag **before closing of </body>** tag by following way-

<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js" integrity="sha384-C6RzsynM9kWDrMNeT87bh95OGNyZPhcTNXj1NW7RuBCsyN/o0jlpcV8Qyq46cDfL" crossorigin="anonymous"></script>

Note: we can copy the code of above <script> tag from www.getbootstrap.com

Limitation of BootstrapCDN: To use BootstrapCDN, we must always connect to the internet.

2) **Offline (By downloading CSS and JavaScript bootstrap library):**

   We can use the bootstrap library to develop our web site by downloading precompiled CSS and JavaScript files from the website https://www.getbootstrap.com/ .

- To use downloaded precompiled CSS files, we use CSS file in <link> tag inside <head> tag by following way:

  <link rel="stylesheet" type="text/css" href="bootstrap.css">

  In the above example, bootstrap.css is the precompiled CSS file that we downloaded from www.getbootstrap.com site.

- To use downloaded precompiled Javascript files, we use javascript file in <script> tag **before closing of </body>** tag by following way:

  <script type="text/javascript" src="bootstrap.js"> </script>

  In the above example, bootstrap.js is the precompiled Javascript file that we downloaded from www.getbootstrap.com site.

Advantages of using downloaded bootstrap CSS and JavaScript: To use Bootstrap offline, there is no need to always connect to the internet.

Note: Downloaded bootstrap.css and bootstrap.js files must be stored inside our web page development folder.

Bootstrap uses:

HTML5 doctype

Bootstrap requires the use of the HTML5 doctype. Without it, you'll see some funky and incomplete styling. To code bootstrap HTML5, it has following structure-

```
<!doctype html>
<html lang="en">
 ...
</html>
```

Viewport <meta>

We know that Bootstrap developed ***mobile first***, a strategy in which we optimize code for mobile devices first and then scale up components as necessary using CSS media queries.

To ensure **proper rendering and touch zooming for all devices**, we add the responsive viewport <meta> tag inside <head> tag as.

```
<meta name="viewport" content="width=device-width, initial-scale=1">
```

First Bootstrap program using BootstrapCDN:

```
<!doctype  html>
<html lang="en">
  <head>
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1">
    <title>Online Bootstrap demo</title>
    <link href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/css/bootstrap.min.css"
rel="stylesheet" integrity="sha384-
T3c6CoIi6uLrA9TneNEoa7RxnatzjcDSCmG1MXxSR1GAsXEV/Dwwykc2MPK8M2HN"
crossorigin="anonymous">
  </head>
  <body>
    <h1>Hello, world!</h1>
<script src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.2/dist/js/bootstrap.bundle.min.js"
integrity="sha384-
C6RzsynM9kWDrMNeT87bh95OGNyZPhcTNXj1NW7RuBCsyN/o0jlpcV8Qyq46cDfL"
crossorigin="anonymous"></script>
  </body>
</html>
```

First Bootstrap program using downloaded CSS and Javascript files:

```
<!doctype  html>
<html lang="en">
<head>
        <title>Offline Bootstrap Demo</title>
        <meta   charset="utf-8">
        <meta   name="viewport"   content="width=device-width, initial-scale=1">
        <link    rel="stylesheet"  type="text/css"  href="bootstrap.css">
        </head>
<body>
        <h1>Welcome to BootStrap</h1>
       <script  type="text/javascript"   src="bootstrap.js"></script>
</body>
</html>
```

Note:

1) In the above example, we use downloaded bootstrap.css and bootstrap.js files in our program. Which are stored locally inside our working web page directory.

2) As per our requirement we can also use bootstrap.min.css or bootstrap.bundle.js or bootstrap.bundle.min.js etc. from the bootstrap framework.

Practical work:

1) **Layout:** **Containers are the most basic layout element in Bootstrap and are required when using our default grid system (Web page is divided into rows and columns). Containers are used to contain, pad, and (sometimes) center the content within them. While containers can be nested, most layouts do not require a nested container.**

Bootstrap comes with three different containers:

- `.container`, which sets a `max-width` at each responsive breakpoint
- `.container-{breakpoint}`, which is `width: 100%` until the specified breakpoint
- `.container-fluid`, which is `width: 100%` (Full width) at all breakpoints

breakpoints (Breakpoints are customizable widths that determine how your responsive website layout behaves across different device or viewport Bootstrap)

Available breakpoints

Bootstrap includes six default breakpoints, sometimes referred to as *grid tiers*, for building responsively. These breakpoints can be customized if you're using our source Sass files.

Breakpoint	Class infix	Dimensions
Extra small	*None*	<576px
Small	`sm`	≥576px
Medium	`md`	≥768px
Large	`lg`	≥992px
Extra large	`xl`	≥1200px
Extra extra large	`xxl`	≥1400px

The table below illustrates how each container's `max-width` compares to the original `.container` and `.container-fluid` across each breakpoint.

	Extra small <576px	Small ≥576px	Medium ≥768px	Large ≥992px	X-Large ≥1200px	XX-Large ≥1400px
`.container`	100%	540px	720px	960px	1140px	1320px
`.container-sm`	100%	540px	720px	960px	1140px	1320px
`.container-md`	100%	100%	720px	960px	1140px	1320px
`.container-lg`	100%	100%	100%	960px	1140px	1320px
`.container-xl`	100%	100%	100%	100%	1140px	1320px
`.container-xxl`	100%	100%	100%	100%	100%	1320px
`.container-fluid`	100%	100%	100%	100%	100%	100%

2) **Nesting of layouts.**
3) **customCSS in bootstrap: Use our created .css file in the program.**
4) **Bootstrap grid. (dividing web page into rows and columns)**

Pract-1) By default bootstrap webpage has 12 max columns. show practically.

(hint: use <div class= "row"> and <div class= "col-1"> etc.)

- We can merge columns as we require. show by practically into next row

Pract-2)

- add any content i.e. text, any image, any html form component, html page using <iframe> tag etc. at each column as per requirement.

- take <marquee> tag in column

- take any html page using <iframe> tag in column

- take <button> control in column

- take image using <img> tag in column

5) **Auto layout columns show practically.**

(Hint: use <div class= "row">

<div class= "col"> first</div>

<div class= "col">Second</div>

</div>)

- take <marquee> tag in column

- take any html page using <iframe> tag in column

- take <button> control in column

- take image using <img> tag in column

Bootstrap Margin and Bootstrap Padding:

-Margin and Padding basically used for space management between HTML elements that we use while creating web pages.

Margin- (m) setting **outer space** for specific HTML element specially for <div> tag.

We can use following margins as follow-

1) mx-size: It is used for horizontal margin
2) my-size: It is used for vertical margin
3) m-size: It is used for both horizontal and vertical margin.

Here, for size we can use any one of the following value:

0 - for classes that eliminate the margin or padding by setting it to 0

1 - (by default) for classes that set the margin or padding to $spacer * 0.25

2 - (by default) for classes that set the margin or padding to $spacer * 0.5

3 - (by default) for classes that set the margin or padding to $spacer * 1.0

4 - (by default) for classes that set the margin or padding to $spacer * 1.5

5 - (by default) for classes that set the margin or padding to $spacer * 3

auto - for classes that set the margin to auto

Here, `$spacer` is a map variable and its default value is 1rem i.e. 16px.

Padding-(p) setting **inner space** for specific HTML element specially for <div> tag.

We can use following padding as follow-

1) px-size: It is used for horizontal padding
2) py-size: It is used for vertical padding
3) p-size: It is used for both horizontal and vertical padding.

Here, for size we can use any one of the following value:

0 - for classes that eliminate the margin or padding by setting it to 0

1 - (by default) for classes that set the margin or padding to $spacer *0 .25

2 - (by default) for classes that set the margin or padding to $spacer * 0.5

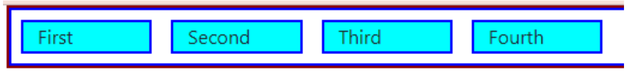
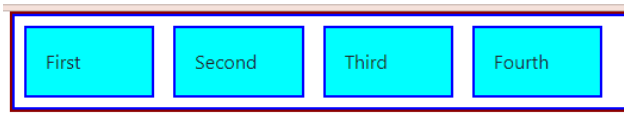
3 - (by default) for classes that set the margin or padding to $spacer * 1.0

4 - (by default) for classes that set the margin or padding to $spacer * 1.5

5 - (by default) for classes that set the margin or padding to $spacer * 3

auto - for classes that set the padding to auto

Following shows use of margin , padding:

Before Margin and Padding output is:	
After setting margin (m-2) output is:	
After setting margin (m-2) and padding (p-3) output is:	

Note: Margin and padding should be applied for column class only.

Practical: Show practically margin and padding

Gutter-(g) :

Gutters are the gaps between column content, created by horizontal padding. We set padding-right and padding-left on each column, and use negative margin to offset that at the start and end of each row to align content.

Gutters start at 1.5rem (24px) wide. This allows us to match our grid to the padding and margin spacers scale.

Gutters can be responsively adjusted.

We can use breakpoint-specific gutter classes to modify horizontal gutters, vertical gutters, and all gutters.

We can use following gutters as follow-

1) gx-size: It is used for horizontal gutter
2) gy-size: It is used for vertical gutter
3) g-size: It is used for both horizontal and vertical gutters.

Here, for size we can use any one of the following value:

0 - for classes that eliminate the gutter by setting it to 0

1 - (by default) for classes that set the gutter to $spacer * 0.25

2 - (by default) for classes that set the gutter to $spacer * 0.5

3 - (by default) for classes that set the gutter to $spacer * 1.0

4 - (by default) for classes that set the gutter to $spacer * 1.5

5 - (by default) for classes that set the gutter to $spacer * 3

auto - for classes that set the gutter to auto

Types of Gutters:

1) Horizontal Gutter:
2) Vertical Gutter:

Note: Gutter should be applied for row class only.

Following shows use of gutters:

Before gutter output is:	
After setting horizontal gutter (gx-3) output is:	

Responsive Images in Bootstrap:

Responsive images are those <u>images that they never become larger than their parent elements and add lightweight styles to them</u>.

In bootstrap images are made responsive with the help of built in class .img-fluid class. The .img-fluid class apply max-width: 100 % and height: auto to the image.

If we apply class .img-fluid to the image then it becomes responsive with respect to its parent element.

Image thumbnails:

In addition to our border-radius utilities, we can use .img-thumbnail to give an image a rounded 1px border appearance.

Display image in rounded shape:

To display images in rounded shape, .rounded-circle bootstrap class is used.

Aligning images:

Images can be aligned at the left or right or middle of the web page.

Images can be aligned with the helper float or text alignment classes.

Whereas block-level images can be centered using the .mx-auto margin utility class.

Aligning image to Right:

To align image at the right side of web page we can use built in css class .float-end

Aligning image to Left:

By default all images will appear at left side but still To align image at the left side of web page we can use built in css class .float-start

Aligning image to Center OR Middle:

To align image at the center of web page we can use built in css class

.mx-auto .d-block

Making image corners rounded:

To make image corners rounded, bootstrap .rounded class is used.

Practical)

Take an image and make it responsive using built-in bootstrap classes. (Hind: Use <img src= "bird.jpg" class= "img-fluid rounded img-thumbnail float-end">

inside container code.

Bootstrap Typography:

Typography is a feature of Bootstrap which is used for styling and formatting the text content. It is used to create customized headings, inline subheadings, lists, paragraphs, aligning, adding more design-oriented font styles, and much more. Bootstrap support global settings for the font stack, Headings and Link styles to be used in the web app for every type of OS and Device to deliver the best user interface.

Typography can be used to create:

Headings

Subheadings

Text and Paragraph font color, font type, and alignment

Lists

Other inline elements

Let's see some of the built-in classes of bootstrap which are used in typography:

- **h1 – h6: To match the font styling of a heading but cannot use the associated HTML element.**
- **text-muted: text-muted fades the text. i.e. text grayed out.**
- **.text-primary: Text styled with class "text-primary" in faint skyblue color**
- **.text-success Text styled with class "text-success" in green color**

- ❖ .text-info   Text styled with class "text-info" in dark skyblue color
- ❖ .text-warning   Text styled with class "text-warning" in orange color
- ❖ .text-danger   Text styled with class "text-danger" in red color
- ❖ .bg-primary   Text background OR Table cell is styled with class "bg-primary" (faint skyblue color)
- ❖ .bg-success   Text background OR Table cell is styled with class "bg-success" (green color)
- ❖ .bg-info   Text background OR Table cell is styled with class "bg-info" (dark skyblue color)
- ❖ .bg-warning   Text background OR Table cell is styled with class "bg-warning" (orange color)
- ❖ .bg-danger   Text background OR Table cell is styled with class "bg-danger" (red color)
- ❖ display: It is used to create better headings.
- ❖ lead: It is used to make a paragraph stand out i.e. Visually better.
- ❖ mark: It is used to highlight the text.
- ❖ small: It is used to create secondary subheadings.
- ❖ initialism: It is used to render abbreviations in slightly small text sizes.
- ❖ text-center: It is used to align the text to the center.
- ❖ text-right: Depicts right-aligned text.
- ❖ text-left: Depicts left-aligned text.
- ❖ text-uppercase: It is used to transform text to uppercase.
- ❖ text-lowercase: It is used to transform text to lowercase.
- ❖ text-capitalize: It is used to transform text to capitalize the first letter of each word leaving other letters in lowercase.
- ❖ list-unstyled: the default list-style and left margin on list items are removed.
- ❖ list-inline: It is used to make the element of the list inline.

Practical) Design web page that uses bootstrap typography.

Hint: <p   class= “h1”>Welcome to Typography</p>

<p   class= “h2”>Sangola College</p>

Use such bootstrap typography classes.

Bootstrap Tables:

We know that tables are more suitable to represent the data or information in rows and column format. For that, HTML provides us a <table> tag with its sub tags.

To design a table in a more attractive, responsive, decent manner, bootstrap has some built-in table classes which are explained as follows.

Lets see these bootstrap table classes in details-

.table class:

Entire table coding is embedded in basic <table> tag with .table bootstrap class

Eg. <table class= "table">

For table heading, use tag <thead> …. </thead>

For table body, use tag <tbody> …. </tbody>

Note that: further <thead> and <tbody> tags may contain <th> or <tr> tag as per requirement.

.table-dark class:

This bootstrap class inverts the colors—with light text on dark (black) backgrounds.

Eg. <table class= "table table-dark">

.table-light class:

This bootstrap class inverts the colors—with text on a light gray background.

Eg. <table class= "table table-light">

.table-striped class:

This bootstrap class add zebra-striping (alternative white and light gray background) to table rows within the <tbody>

Eg. <table class= "table table-striped">

.table-bordered class:

This bootstrap class adds borders to all table content.

Eg. <table class= "table table-bordered">

.table-hover class:

This bootstrap class enables a hover state on table rows within a <tbody> tab.

Eg. <table class= "table table-hover">

.table-sm class:

This bootstrap class makes tables more compact by cutting cell padding in half size.

Eg. <table class= "table table-sm">

Note: we can use all these bootstrap classes as combination as per requirements like <table class="table table-striped table-dark table-bordered">

Contextual bootstrap classes:

The contextual bootstrap classes are used to color table rows or individual cells.

The bootstrap contextual classes are-

- **.table-active**
- **.table-primary**
- **.table-secondary**
- **.table-success**
- **.table-danger**
- **.table-warning**
- **.table-info**
- **.table-light**
- **.table-dark**

To make entire row colorable, use any one of the above class with <tr> tag as follow-

Eg. <tr class= “table-danger”>...</tr>

To make specific cell of table colorable, use any one of the above class with <td> or <th> tag as follow-

Eg. <td class= “table-primary”>...</td>

<th class= “table-success”>...</th>

(First Implement Table without bootstrap and then use built-in bootstrap classes to Table and show the difference for students.)

Practical 1) Design any table and apply all above bootstrap table classes to it.

DemoExample-

```html
<div class="container">
    <table class="table-striped table-bordered table-hover table-sm">
<thead>
    <tr class="table-success">
        <th>Sr.No</th><th>Name</th><th>City</th>
    </tr>
</thead>
<tbody>
    <tr>
        <th class="table-danger">1.</th><th class="table-warning">AMOL</th><th class="table-primary">SANGOLA</th>
    </tr>
    <tr>
        <th class="table-primary">2.</th><th class="table-info">KIRAN</th><th class="table-primary">SOLAPUR</th>
    </tr>
</tbody>
    </table>
</div>
```

The output of above HTML code is:

Sr.No	Name	City
1.	AMOL	SANGOLA
2.	KIRAN	SOLAPUR

Bootstrap Forms:

We know that an HTML Form is used to collect data or information or any input from a user (client) . Then this collected data is most often sent to a server for processing.

The bootstrap framework also has built-in form classes which are used to display form more attractive, more decent, more responsive. Lets see these bootstrap form classes in details-

.form-group class:

This is a basic bootstrap form class which is used to make a group of all form controls. This class wraps labels and form controls in <div class="form-group"> (needed for optimum spacing)

Now add class .form-control class to all textual <input>, <textarea>, and <select> elements then they will add in the form-group.

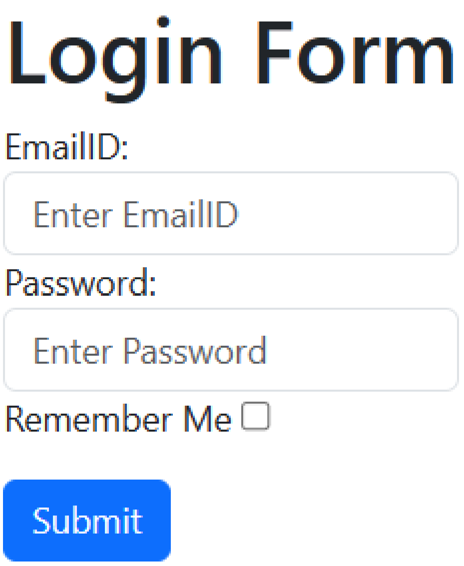
Following program creates a simple Bootstrap form with two input fields email and password with one checkbox, and a submit button:

Note: Following program should use bootstrap.css and mystyle.css in <link> tag.

(First Implement Form without bootstrap and then use built-in bootstrap classes to Form and show the difference for students.)

```html
<div class="container">
<h1>Login Form</h1>
<form class="form-group">
<div class="row">
    <div class="col-auto">
        <label for="mail">EmailID:</label>
        <input type="email" id="mail" placeholder="Enter EmailID" class="form-control">
    </div>
</div>
<div class="row">
    <div class="col-auto">
        <label for="pwd">Password:</label>
        <input type="password" id="pwd" placeholder="Enter Password" class="form-control">
    </div>
</div>
<div class="row">
    <div class="col-auto">
        <label for="chk">Remember Me</label>
        <input type="checkbox" id="chk">
    </div>
</div>
<div class="row">
    <div class="col-auto">
        <button class="btn btn-primary my-3">Submit</button>
    </div>
</div>
</form>
</div>
```

The output of above HTML code is-



`.form-control-lg and .form-control-sm :`

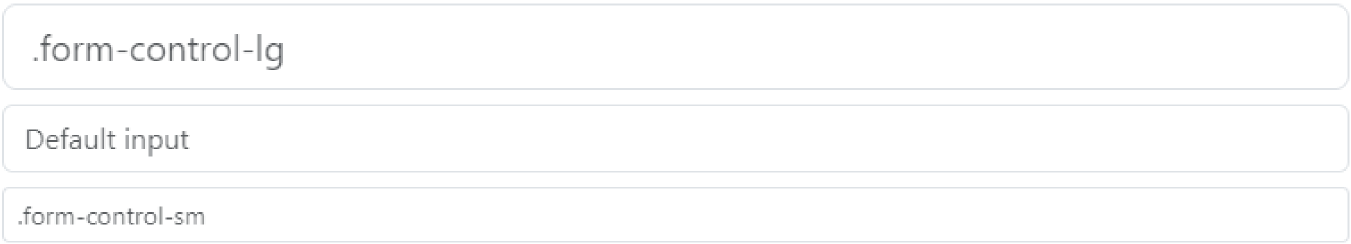
Using .form-control-lg and .form-control-sm bootstrap classes we can manage the height of <input> element.

Eg.

```html
<input class="form-control form-control-lg" type="text" placeholder=".form-
```

```
control-lg">
<input class="form-control" type="text" placeholder="Default input" >
<input class="form-control form-control-sm" type="text" placeholder=".form-control-sm">
```

The output of above HTML code is:



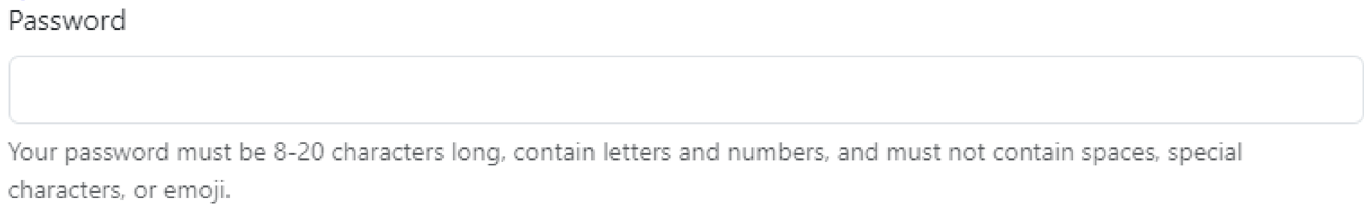
.form-text class:

Block-level or inline-level form text can be created using .form-text class.

Form text <u>below inputs</u> can be styled with .form-text.

.form-text class sets a top margin for easy spacing from the inputs above.

Eg.



The HTML code of above output:

```
<label for="inputPassword5">Password</label>
<input type="password" id="inputPassword5" class="form-control">
<div id="passwordHelpBlock" class="form-text">
  Your password must be 8-20 characters long, contain letters and numbers, and must not contain spaces, special characters, or emoji.
</div>
```

Bootstrap Form Controls:

To design a form, we can take different bootstrap form controls such as <input>, <select>, and <textarea> which are styled with the .form-control class.

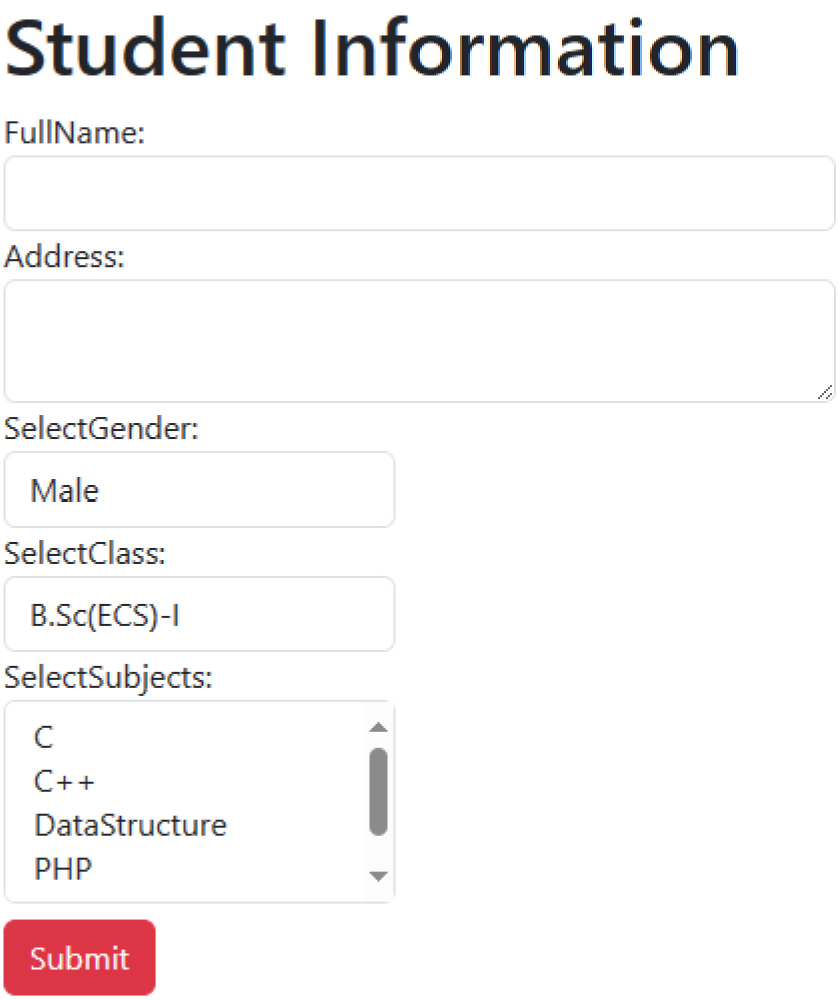
The use of <input>, <select> and <textarea> controls are as follow-

Note: Following program should use bootstrap.css and mystyle.css in <link> tag.

```html
<div class="container">
<h1>Student Information</h1>
<form class="form-group">
	<div class="row">
		<div class="col-4">
			<label for="sname">FullName:</label>
			<input type="text" id="sname" class="form-control">
		</div>
	</div>
	<div class="row">
		<div class="col-4">
			<label for="add">Address:</label>
			<textarea id="add" class="form-control"></textarea>
		</div>
	</div>
	<div class="row">
		<div class="col-2">
			<label for="gen">SelectGender:</label>
			<select id="gen" class="form-control">
				<option>Male</option>
				<option>Female</option>
			</select>
		</div>
	</div>
	<div class="row">
		<div class="col-2">
			<label for="cls">SelectClass:</label>
			<select id="cls" class="form-control">
				<option>B.Sc(ECS)-I</option>
				<option>B.Sc(ECS)-II</option>
				<option>B.Sc(ECS)-III</option>
			</select>
		</div>
	</div>
<div class="row">
		<div class="col-2">
			<label for="sub">SelectSubjects:</label>
			<select multiple  id="sub"  class="form-control">
				<option>C</option>
				<option>C++</option>
				<option>DataStructure</option>
				<option>PHP</option>
				<option>Java</option>
```

```
                    </select>
              </div>
</div>
<div class="row">
       <div class="col-2">
              <button class="btn btn-danger   my-2">Submit</button>
       </div>
</div>
</form>
</div>
```

The output of above HTML code is:



File input control: (File selection Control)

1) For input or upload a file first we have to select it, for that we use <input> tag and set its attribute type= "file" also to make input file control more descent, we use bootstrap classes .form-control-file and .form-control

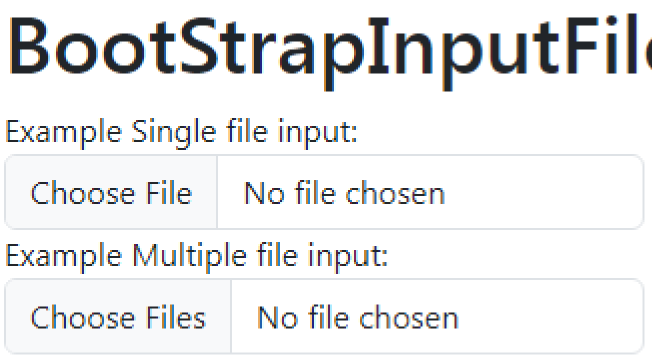
2) For input or upload multiple files, we use <input> tag and set its attribute type= "file" and also add multiple property to <input> tag.

In bootstrap, we can input/upload single as well as multiple files using file input control as follow-

Note: Following program should use bootstrap.css and mystyle.css in <link> tag.

```
<div class="container">
<h1>BootStrapInputFiles</h1>
  <form class="form-group">
    <div class="row">
                <div class="col-auto">
                        <label for="ipSfile">Example Single file input: </label>
                        <input type="file" class="form-control-file  form-control"
id="ipSfile">
                </div>
        </div>
        <div class="row">
                <div class="col-auto">
                        <label for="ipMfile">Example Multiple file input: </label>
                        <input type="file" class="form-control-file  form-control"
id="ipMfile"    multiple>
                </div>
        </div>
    </form>
</div>
```

The output of above HTML code is:



Date input control:

For input a date, for that we use <input> tag and set its attribute type= "date" also to make that input control more descent, we use bootstrap class .form-control.

Following HTML code shows use of date input control:

```
<div class="container">
        <h3>DateInputDemo</h3>
```

```
        <form class="form-group">
                <div class="row  col-2">
                        <label id="dt">DateOfBirth:</label>
                        <input type="date" id="dt"  class="form-control">
                </div>
        </form>
</div>
```

The output is:

Checkbox and Radio controls:

Default checkboxes and radios are improved in bootstrap using .form-check and .form-check-input classes. The .form-check class sets proper margin and padding whereas .form-check-input class improves the layout and behavior of their checkbox and radios.

Note: 1) Checkboxes are for selecting one or several options in a list, while radios are for selecting one option from many.

2) To select only one option from multiple radios, set the name attribute with same name of all radioes.

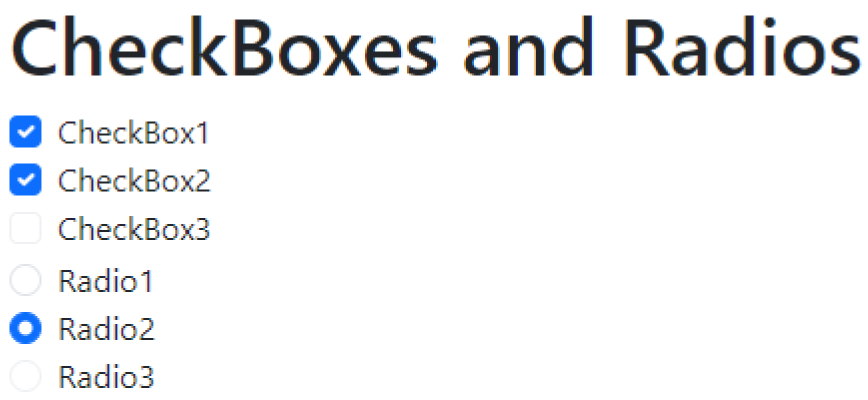
3) We can make checkboxes and radios disabled by setting the attribute disabled of the respective <input> tag of checkbox and radio.

Note: Following program should use bootstrap.css and mystyle.css in <link> tag.

```
<div class="container">
        <h1>CheckBoxes and Radios</h1>
 <form class="form-group">
        <div class=" row   form-check">
                <label for="ch1"><input type="checkbox" name="" class="form-
check-input" id="ch1"> CheckBox1</label>
                <label for="ch2"><input type="checkbox" name="" class="form-
check-input" id="ch2"> CheckBox2</label>
                <label for="ch3"><input type="checkbox" name="" class="form-
check-input" id="ch3"  disabled> CheckBox3</label>
        </div>
        <div class=" row   form-check">
```

```
                <label for="op1"><input type="radio" name="nm" class="form-check-input" id="op1"> Radio1</label>
                <label for="op2"><input type="radio" name="nm" class="form-check-input" id="op2"> Radio2</label>
                <label for="op3"><input type="radio" name="nm" class="form-check-input" id="op3"  disabled> Radio3</label>
        </div>
  </form>
</div>
```

The output of above HTML code:



.form-check-inline and .check-input-inline classes:

We can make multiple checkboxes and multiple radios in line i.e. they will appear in a single row by using .form-check-inline and .check-input-inline classes.

Following program shows use of .form-check-inline and .check-input-inline

```
<div class="container">
        <h1>Inline CheckBoxes and Radios</h1>
        <form class="form-group">
        <div class="form-check-inline">
                <label>Select Subjects: </label>
                <label for="ch1"><input type="checkbox" name="" id="ch1" class="form-check-input"> C++</label>
                <label for="ch2"><input type="checkbox" name="" id="ch2" class="form-check-input"> DataStructure</label>
        </div>
        <div class="form-check-inline" >
                <label>Select Gender: </label>
                <label for="op1"><input type="radio" name="gen" id="op1" class="form-check-input"> Male</label>
                <label for="op2"><input type="radio" name="gen" id="op2" class="form-check-input"> Female</label>
        </div>
```

```
        </form>
</div>
```

The output of above HTML code:

Button control:

Button control is one of the important control of form. Basically it is used to perform actions in forms.

Bootstrap provides custom button styles for actions in forms, dialogs, and more with support for multiple sizes, states, and more.

Bootstrap includes several predefined button styles, each serving its own semantic purpose.

Note that, we can add button control on form using <button> tag with type= “button” as follow-

```
<button    type="button">ClickMe</button>
```

.btn class:

This is a basic bootstrap button class that displays buttons in a more decent format.

Also, bootstrap provides more button classes to make .btn class colorable. Which are discussed as follows-

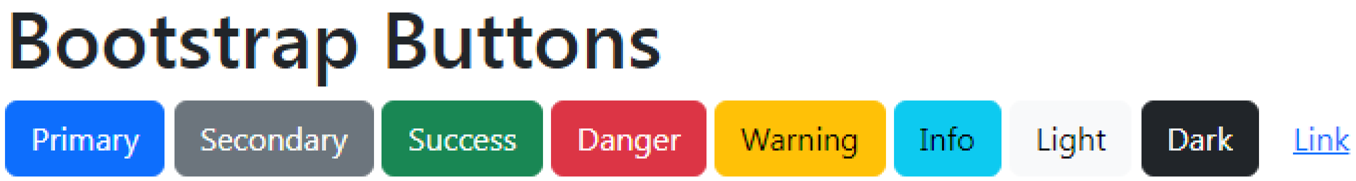
Bootstrap button class	Use
btn-primary	Display blue coloured button
btn-secondary	Display dark gray coloured button
btn-success	Display green coloured button
btn-danger	Display red coloured button
btn-warning	Display orange coloured button
btn-info	Display cyan coloured button
btn-light	Display light gray coloured button
btn-dark	Display black coloured button
btn-link	Display button like html link

Following program shows use of different bootstrap button classes-

Note: Following program should use bootstrap.css and mystyle.css in <link> tag.

```
<div class="container">
<h1>Bootstrap Buttons</h1>
<form class="form-group">
<button type="button" class="btn btn-primary">Primary</button>
<button type="button" class="btn btn-secondary">Secondary</button>
<button type="button" class="btn btn-success">Success</button>
<button type="button" class="btn btn-danger">Danger</button>
<button type="button" class="btn btn-warning">Warning</button>
<button type="button" class="btn btn-info">Info</button>
<button type="button" class="btn btn-light">Light</button>
<button type="button" class="btn btn-dark">Dark</button>
<button type="button" class="btn btn-link">Link</button>
 </form>
</div>
```

The output of above HTML code is-



.btn-outline-* class:

These bootstrap classes are used to remove all background images and colors on any button. It just places button text and button border with respective colors.

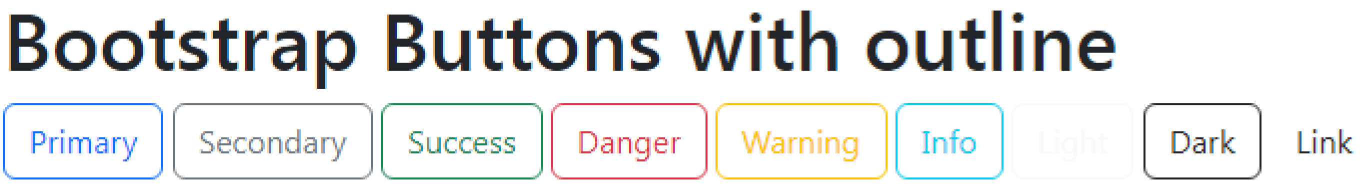
Note: In .btn-outline-* class, * is replaced by primary, secondary, success, danger etc. as per requirement.

Following program shows use of .btn-outline-* class

```
<div class="container">
<h1>Bootstrap Buttons with outline</h1>
<form class="form-group">
<button type="button" class="btn btn-outline-primary">Primary</button>
<button type="button" class="btn btn-outline-
secondary">Secondary</button>
<button type="button" class="btn btn-outline-success">Success</button>
<button type="button" class="btn btn-outline-danger">Danger</button>
<button type="button" class="btn btn-outline-warning">Warning</button>
<button type="button" class="btn btn-outline-info">Info</button>
<button type="button" class="btn btn-outline-light">Light</button>
<button type="button" class="btn btn-outline-dark">Dark</button>
<button type="button" class="btn btn-outline-link">Link</button>
 </form>
```

```
</div>
```

The output of above HTML code is:



.btn-lg and btn-sm class:

The .btn-lg bootstrap class display button with large size.

The .btn-sm bootstrap class display button with small size.

Bootstrap drop-down:

Bootstrap dropdowns are toggle (a suitable selection between multiple options) contextual overlays for displaying lists of links.

Bootstrap dropdowns, made interactive with the included Bootstrap dropdown JavaScript plugin. They're toggled by clicking.

Dropdowns are built on a third party library named Popper, which provides dynamic positioning and viewport detection. Be sure to include popper.min.js before Bootstrap's JavaScript or use bootstrap.bundle.min.js or bootstrap.bundle.js which contains Popper.

Note: Popper isn't used to position dropdowns in navbars though as dynamic positioning isn't required.

There are different format of dropdowns in bootstrap, we explained as follow-

Single button:

Any single .btn can be turned into a dropdown toggle with some markup changes. Here's how you can put them to work with <button> elements:

To make a button as a dropdown, .dropdown-toggle class is used. Also we have to set its attribute data-bs-toggle="dropdown" of the button.

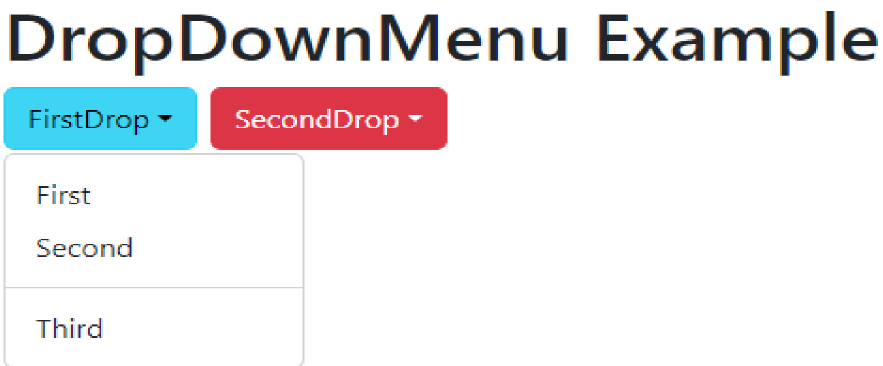
Also, to make menus in dropdown, we use <ul> tag having class attribute .dropdown-menu and <li> tag with <a> having class attribute .dropdown-item. which is explained as follow-

```
<ul class="dropdown-menu">
        <li><a class="dropdown-item" href="#">NewFile</a></li>
        <li><a class="dropdown-item" href="#">OpenFile</a></li>
        <li><a class="dropdown-item" href="#">CloseFile</a></li>
```


Following program shows displaying single button dropdown in bootstrap-

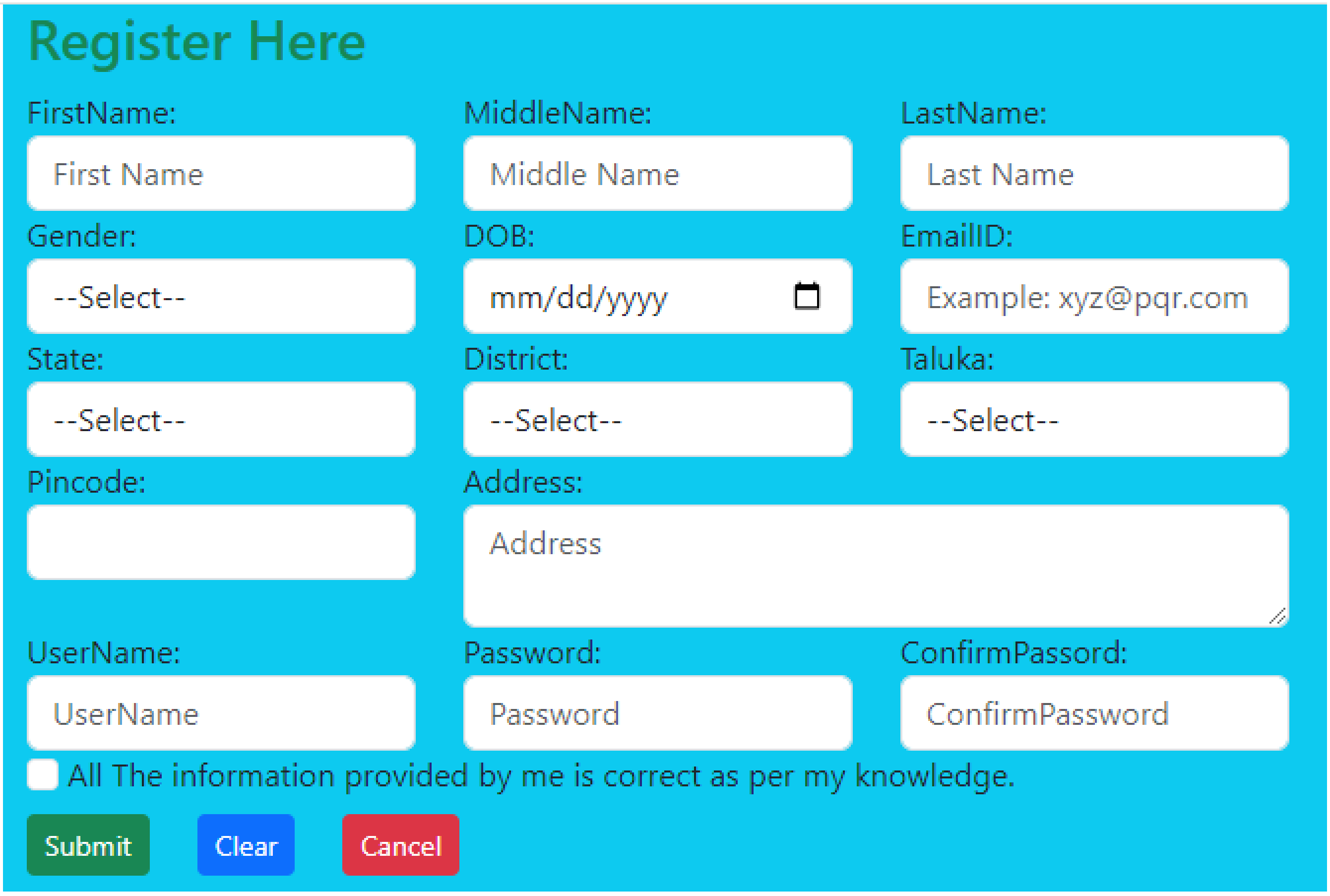
```
<body>
<div class="container">
        <h1>DropDownMenu Example</h1>
        <div class="row">
                <div class="col-1">
                        <button type="button" class="btn btn-info  dropdown-toggle"
data-bs-toggle="dropdown">FirstDrop</button>
                        <ul class="dropdown-menu">
                                <li><a href="#" class="dropdown-item">First</a></li>
                                <li><a href="#" class="dropdown-
item">Second</a></li>
                                <li><hr class="dropdown-divider"></li>
                                <li><a href="#" class="dropdown-item">Third</a></li>
                        </ul>
                </div>
                <div class="col-1">
                        <button type="button" class="btn   btn-danger  dropdown-
toggle"  data-bs-toggle="dropdown">SecondDrop</button>
                        <ul class="dropdown-menu">
                        <li><a href="#" class="dropdown-item">First</a></li>
                        <li><a href="#" class="dropdown-item">Second</a></li>
                        </ul>
                </div>
        </div>
</div>
<script type="text/javascript" src="bootstrap.bundle.js"></script>
</body>
```

The output of above HTML code is:



Practical Task)

Design a bootstrap HTML form as follow-



Bootstrap Themes:

Bootstrap Themes is a collection of the built-in templates which was created by Bootstrap's creators.

These are the readymade templates. By using this, we can build any website as per customer requirement.

The bootstrap theme minimizes the developer efforts such that the developer designs the website quickly.

Readymade free bootstrap templates i.e. themes are available at:

https://bootstrapmade.com/

Ready-made bootstrap templates i.e. themes are also available at https://themes.getbootstrap.com/ but these are the paid ones.

Practical: Download any free template and update as per requirement and show for students.

Homework: From this site give homework for students to design websites.

Note: Be always connect to internet while working on bootstrap template

Bootstrap components:

Components are necessary to build a website. i.e. website is built with the help of different components.

Bootstrap has a number of built-in components that can be reused to provide a good user experience and user interactions in a web page.

These components are navigation bars, pop-ups, dropdowns, icons, buttons, pre-designed forms and also sizing options for different DOM elements.

These reusable components are built to provide iconography, dropdowns, input groups, navigation, alerts, and much more.

Lets see some of the bootstrap components in details-

(Explain online from https://getbootstrap.com/docs/5.3/components/accordion/)

Note: To work bootstrap componets properly, must use bootstrap.js or bootstrap.bundle.js or its related javascript file in the program using <script> tag). Since, it contains JavaScript plugins.

JavaScript with bootstrap: